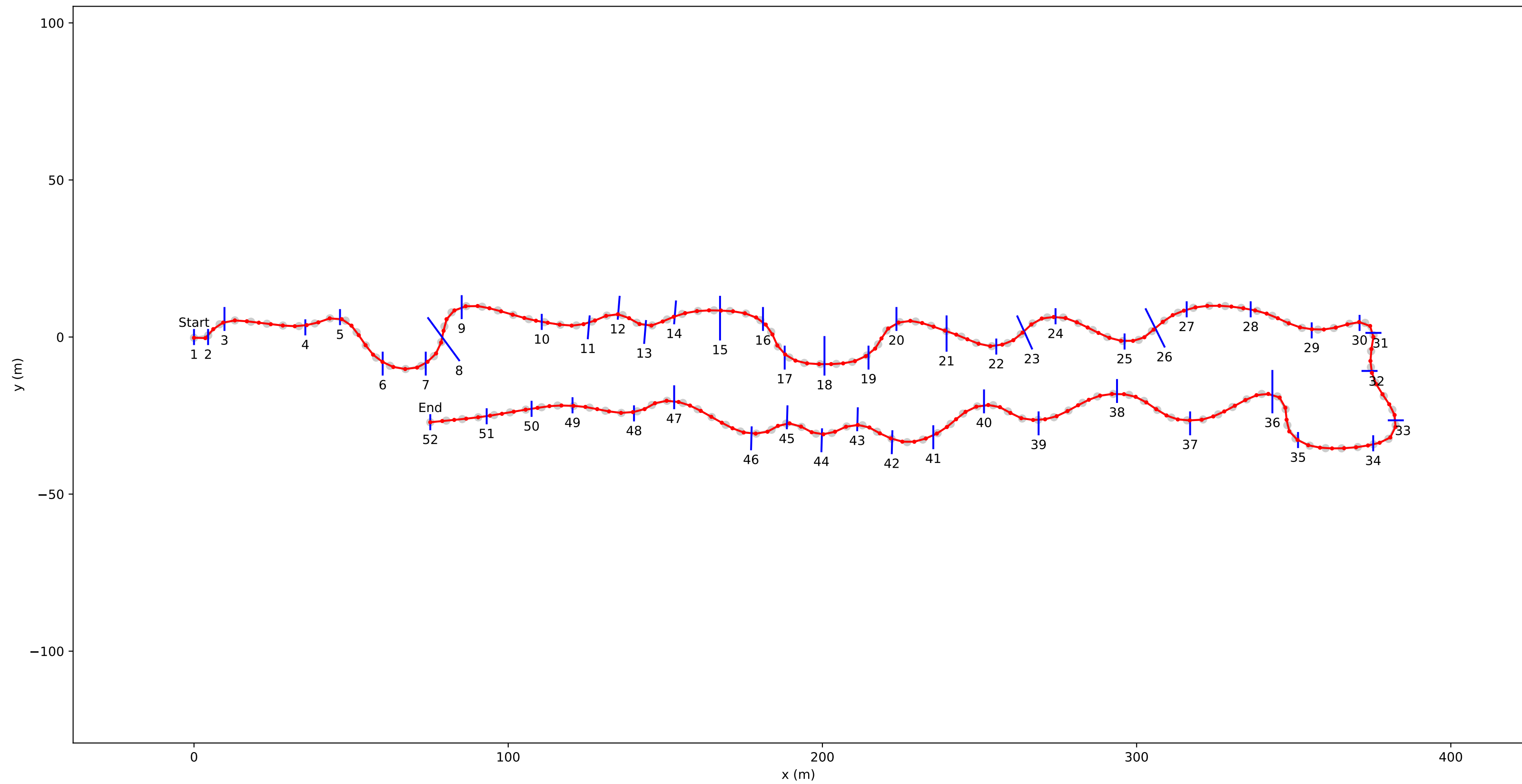


Map plot: optimized path through gates



Track Generation Report

=====
Generated: 2026-05-31 18:45:24
Gate file: ref_gates_autoX.csv
Gate count: 52
Saved DOFs file: opt_dofs_2026-05-31_18-45-23.npy

Optimization parameters
 method: trust-constr
 maxiter: 500
 ftol: 1e-07

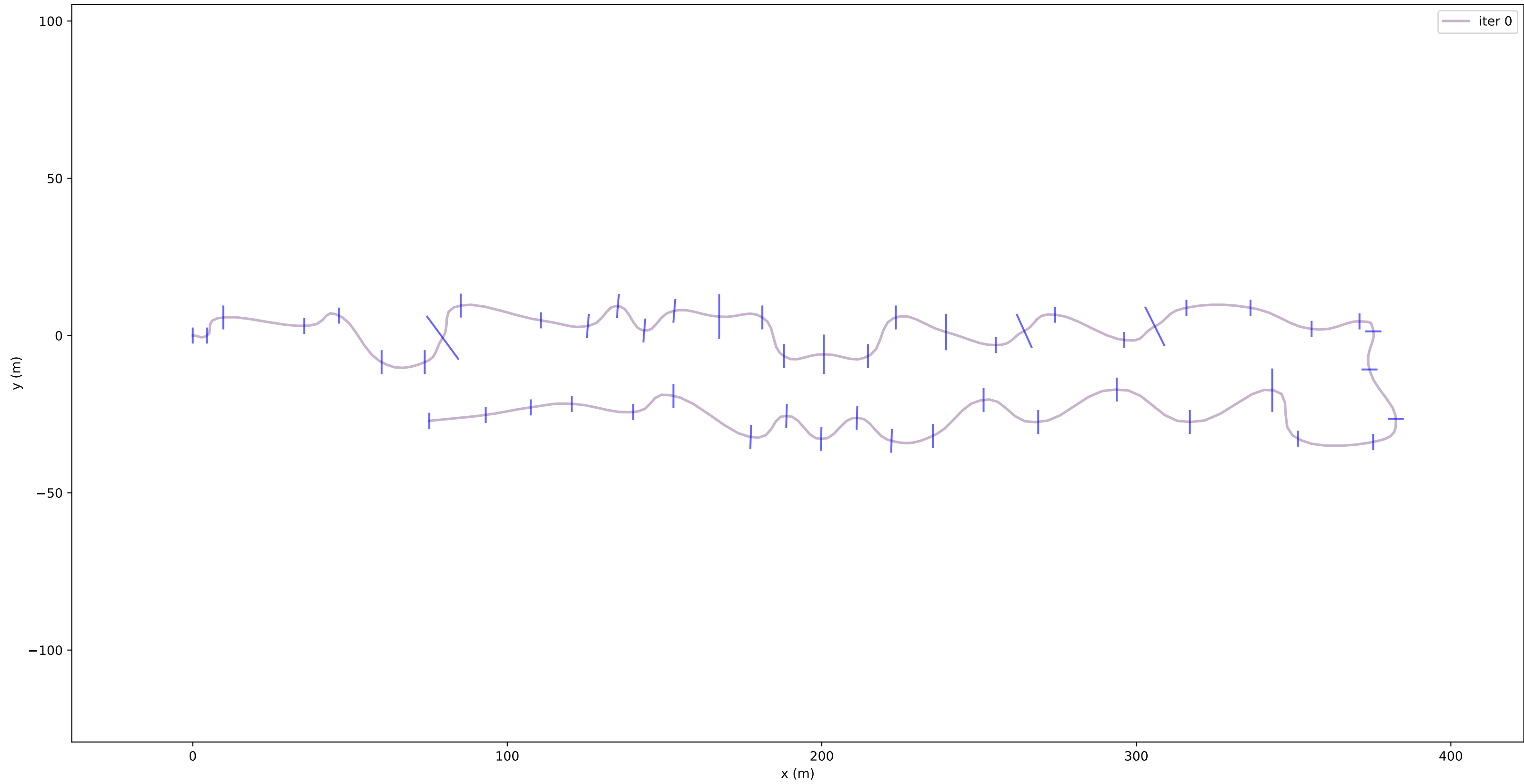
Point counts
 objective function points: 208
 path generation points: 156

Results
 total optimization time: 18964.949 s
 total path length: 792.2865 m

total objective function value: 4.457373
Per-step time estimate (one measured objective evaluation)
 make spline from dofs: 10.76 ms
 path length function: 0.53 ms
 equidistant times generated: 0.36 ms
 curvature objective function: 274.34 ms

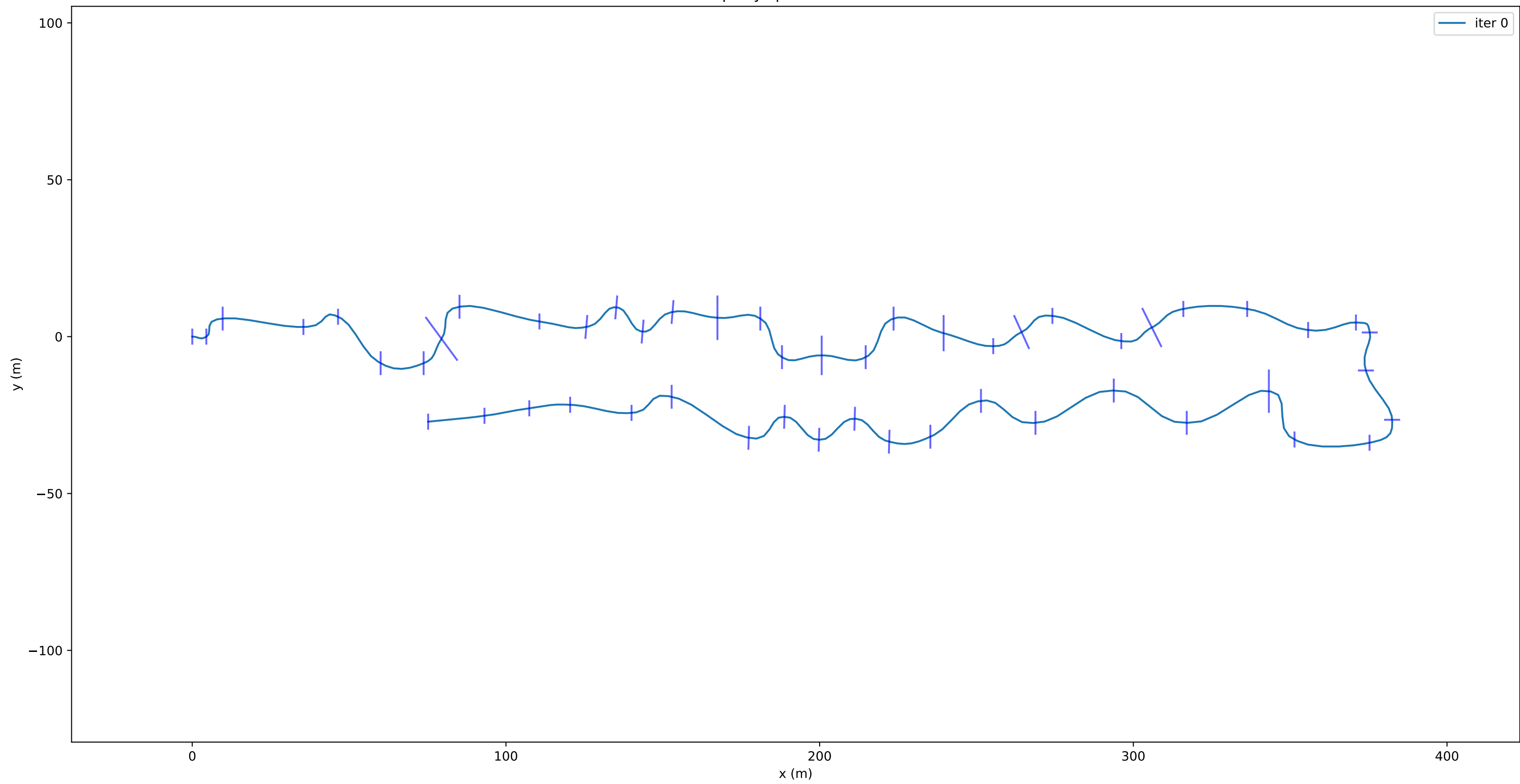
Optimization progression

Optimization progression (1 iterations)



Equally spaced iterations

1 equally spaced iterations



Instant radius vs distance

